Introduction to Photosynthesis

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy, usually from the sun, into chemical energy stored in glucose. This process sustains nearly all life on Earth by producing the oxygen we breathe and the organic compounds that form the base of most food chains.

The Overall Equation

The overall chemical equation for photosynthesis is: six molecules of carbon dioxide plus six molecules of water, in the presence of light energy, yield one molecule of glucose plus six molecules of oxygen. In short: $6\ CO_2 + 6\ H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6\ O_2$.

The Two Main Stages

Photosynthesis occurs in two linked stages inside the chloroplast. The light-dependent reactions take place in the thylakoid membrane, where chlorophyll absorbs sunlight and uses that energy to split water molecules, releasing oxygen and producing ATP and NADPH. The light-independent reactions, also called the Calvin cycle, take place in the stroma. They use the ATP and NADPH from the first stage to fix carbon dioxide into glucose.

Why Chlorophyll Matters

Chlorophyll is the green pigment found in chloroplasts. It absorbs mostly red and blue wavelengths of light while reflecting green light, which is why most plant leaves appear green. Without chlorophyll, plants could not capture the light energy needed to drive the light-dependent reactions.

Factors That Affect the Rate of Photosynthesis

Several environmental factors influence how quickly a plant photosynthesizes: light intensity, carbon dioxide concentration, and temperature. Increasing light intensity or carbon dioxide concentration generally increases the rate of photosynthesis up to a point, after which some other factor becomes limiting. Extreme temperatures can denature the enzymes involved in the Calvin cycle, slowing the process down.

Why Photosynthesis Matters

Photosynthesis is the foundation of nearly every food chain on Earth. It also regulates atmospheric oxygen and carbon dioxide levels, making it essential to the planet’s climate and to the survival of aerobic organisms, including humans.